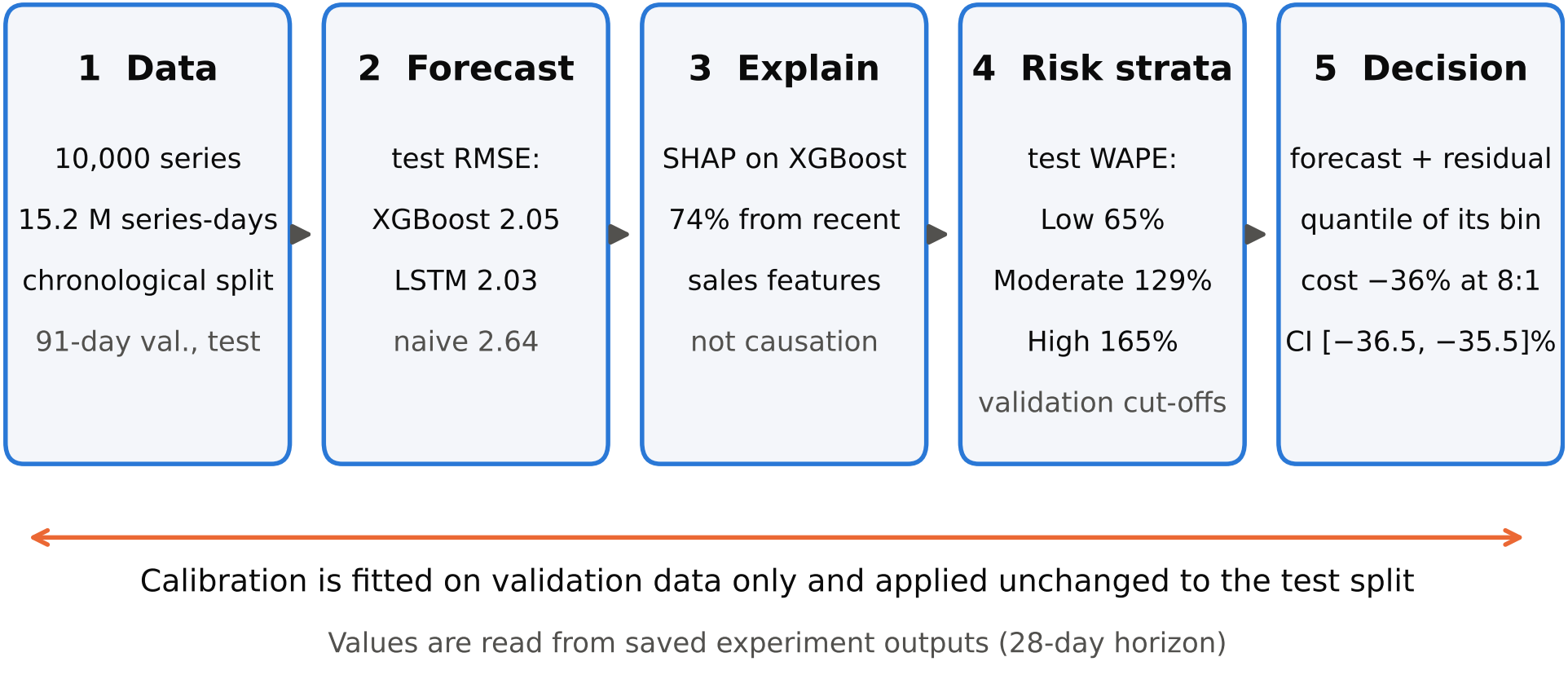
1 Data
10,000 series
15.2 M series-days
chronological split
91-day val., test
2 Forecast
test RMSE:
XGBoost 2.05
LSTM 2.03
naive 2.64
3 Explain
SHAP on XGBoost
74% from recent
sales features
not causation
4 Risk strata
test WAPE:
Low 65%
Moderate 129%
High 165%
validation cut-offs
5 Decision
forecast + residual
quantile of its bin
cost −36% at 8:1
CI [−36.5, −35.5]%
Calibration is fitted on validation data only and applied unchanged to the test split
Values are read from saved experiment outputs (28-day horizon)